

TASK 1:

```
May 2 18:47
muhammad-affan-rasheed@muhammad-affan-rasheed-VirtualBox: ~/Desktop
muhammad-affan-rasheed@muhammad-affan-rasheed-VirtualBox:~/Desktop$ gcc task1.c -o t1
muhammad-affan-rasheed@muhammad-affan-rasheed-VirtualBox:~/Desktop$ cat task1.c
#include <stdio.h>
#include <unistd.h>
#include <signal.h>

void signal_handler(int num){
    printf("I am here because Ctrl+c was pressed\n");
}

int main(){
    struct sigaction sa;
    sa.sa_handler = signal_handler;
    sigemptyset(&sa.sa_mask);
    sa.sa_flags = 0;

    sigaction(SIGINT, &sa, NULL);

    while(1){
        sleep(1);
        printf("Process Running \n");
    }
    return 0;
}
muhammad-affan-rasheed@muhammad-affan-rasheed-VirtualBox:~/Desktop$ ./t1
Process Running
Process Running
^CI am here because Ctrl+c was pressed
Process Running
Process Running
^Z
[1]+  Stopped                  ./t1
muhammad-affan-rasheed@muhammad-affan-rasheed-VirtualBox:~/Desktop$
```

TASK 2:

```
May 2 18:48
muhammad-affan-rasheed@muhammad-affan-rasheed-VirtualBox: ~/Desktop
muhammad-affan-rasheed@muhammad-affan-rasheed-VirtualBox:~/Desktop$ gcc task2.c -o t2
muhammad-affan-rasheed@muhammad-affan-rasheed-VirtualBox:~/Desktop$ cat task2.c
#include <stdio.h>
#include <signal.h>
#include <unistd.h>

int main(){
    signal(SIGKILL, SIG_IGN);
    signal(SIGSTOP, SIG_IGN);

    while(1){
        printf("Trying to ignore SIGKILL and SIGSTOP\n");
        sleep(2);
    }
    return 0;
}
muhammad-affan-rasheed@muhammad-affan-rasheed-VirtualBox:~/Desktop$ ./t2
Trying to ignore SIGKILL and SIGSTOP
Trying to ignore SIGKILL and SIGSTOP
Trying to ignore SIGKILL and SIGSTOP
Trying to ignore SIGKILL and SIGSTOP
Trying to ignore SIGKILL and SIGSTOP
^C
muhammad-affan-rasheed@muhammad-affan-rasheed-VirtualBox:~/Desktop$
```

TASK 3:

```
muhammad-affan-rasheed@muhammad-affan-rasheed-VirtualBox: ~/Desktop
muhammad-affan-rasheed@muhammad-affan-rasheed-VirtualBox:~/Desktop$ gcc task3.c -o t3
muhammad-affan-rasheed@muhammad-affan-rasheed-VirtualBox:~/Desktop$ cat task3.c

#include <stdio.h>
#include <signal.h>
#include <unistd.h>
#include <string.h>

#define SIGCOPY SIGUSR1
#define SIGPASTE SIGUSR2

char msg[100] = "ello guvnor!";
char buf[100];

void copy_handler(int sig) {
    strcpy(buf, msg);
    printf("copied: %s\n", buf);
}

void paste_handler(int sig) {
    strcat(msg, buf);
    printf("after paste: %s\n", msg);
}

int main() {
    signal(SIGCOPY, copy_handler);
    signal(SIGPASTE, paste_handler);

    printf("pid: %d\n", getpid());
    while(1) {
        sleep(1);
    }
    return 0;
}

muhammad-affan-rasheed@muhammad-affan-rasheed-VirtualBox:~/Desktop$ ./t3
pid: 5009
copied: ello guvnor!
after paste: ello guvnor!ello guvnor!
copied: ello guvnor!ello guvnor!
after paste: ello guvnor!ello guvnor!ello guvnor!ello guvnor!
Terminated
muhammad-affan-rasheed@muhammad-affan-rasheed-VirtualBox:~/Desktop$

muhammad-affan-rasheed@muhammad-affan-rasheed-VirtualBox:~/Desktop$ kill -10 5009
muhammad-affan-rasheed@muhammad-affan-rasheed-VirtualBox:~/Desktop$ kill -12 5009
muhammad-affan-rasheed@muhammad-affan-rasheed-VirtualBox:~/Desktop$ kill -10 5009
muhammad-affan-rasheed@muhammad-affan-rasheed-VirtualBox:~/Desktop$ kill -12 5009
muhammad-affan-rasheed@muhammad-affan-rasheed-VirtualBox:~/Desktop$ kill 5009
muhammad-affan-rasheed@muhammad-affan-rasheed-VirtualBox:~/Desktop$ S
```

TASK 4:

```
muhammad-affan-rasheed@muhammad-affan-rasheed-VirtualBox: ~/Desktop
muhammad-affan-rasheed@muhammad-affan-rasheed-VirtualBox:~/Desktop$ gcc task4.c -o t4
muhammad-affan-rasheed@muhammad-affan-rasheed-VirtualBox:~/Desktop$ cat task4.c

#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <signal.h>
#include <time.h>

#define SIGTEMP SIGUSR1

int running = 1;

void temp_handler(int sig){
    printf("warning: crit tmp reached\n");
}

void stop_handler(int sig){
    running = 0;
}

int main(){
    signal(SIGTEMP, temp_handler);
    signal(SIGINT, stop_handler);
    srand(time(NULL));
    while(running == 1){
        int temp = rand() % 101;
        printf("temp: %d celcius\n", temp);
        if(temp > 90){
            raise(SIGTEMP);
        }
        sleep(2);
    }
    printf("monitoring stopped\n");
    return 0;
}

muhammad-affan-rasheed@muhammad-affan-rasheed-VirtualBox:~/Desktop$ ./t4
temp: 73 celcius
temp: 100 celcius
warning: crit tmp reached
temp: 7 celcius
temp: 38 celcius
temp: 24 celcius
temp: 19 celcius
temp: 53 celcius
temp: 45 celcius
*monitoring stopped
muhammad-affan-rasheed@muhammad-affan-rasheed-VirtualBox:~/Desktop$ ./t4
temp: 72 celcius
temp: 18 celcius
temp: 29 celcius
temp: 5 celcius
temp: 62 celcius
temp: 98 celcius
warning: crit tmp reached
temp: 57 celcius
temp: 43 celcius
*monitoring stopped
muhammad-affan-rasheed@muhammad-affan-rasheed-VirtualBox:~/Desktop$
```